```
license: mit
language:
- en
base_model:
- Qwen/Qwen3-8B
- microsoft/phi-4
tags:
- fine-tuning
- lora
- qlora
- dora
- rslora
- reasoning
- instruction-tuning
- ablation-study
- expert-style
library_name: transformers
pipeline_tag: text-generation
```


Fine-Tuning for Expert Voice and Reasoning Style

A Systematic Ablation Study on Input/Output Format, Adapter Recipes, and Training Data Structure

Overview

We fine-tuned a series of open-weight LLMs to reproduce the reasoning and teaching style of domain experts — not to produce topically correct answers, but to generate responses that move the way an expert moves: the structure of thought, the specific quality of engagement.

The central contribution is the **training data unit**: structured multi-component objects we call **reasoning atoms**, each capturing a complete expert reasoning event. The adapter recipe composites five LoRA techniques. The ablation series — 8 format variants, held adapter constant — is what determined the final format.

This document covers the ablation findings, adapter recipe, and reproduction notes. No domain-specific content is included. The method is transferable to any domain with sufficient expert text.

Key Result

> ***The clearest finding from the ablation: eval loss did not reliably track teaching quality.*** *The format with the best training metrics (eval loss 1.496, accuracy 68.7%) produced the weakest qualitative output — generic drift with no expert voice. The format with mediocre metrics (eval loss 1.922) produced the strongest expert-style responses. Qualitative evaluation, not checkpoint loss, is the right selection criterion for stylistic fine-tuning.*

At full scale (22,497 atoms, V6 format, Qwen3-8B): eval loss **1.570**, human evaluation **9.0+ across all 5 categories**, zero generic instructional drift, 9/10 expert voices identifiable.

Cross-architecture validation on Phi-4 14B with the same data and format: eval loss **1.456**, human evaluation **~9.2/10**. The format generalizes.

Artifacts in This Project

All models are GGUF Q8_0 quantized. Compatible with LM Studio, Ollama, llama.cpp, Jan, and similar runtimes.

Profile: https://huggingface.co/Sathman

Combined Models (trained on all expert sources, 22,497 atoms)

Model	Base	Link
Meditation-Agent-8B-GGUF	Qwen3-8B	https://huggingface.co/Sathman/Meditation-Agent-8B-GGUF
Meditation-Agent-Phi4-GGUF	Phi-4 14B	https://huggingface.co/Sathman/Meditation-Agent-Phi4-GGUF

Single-Expert Models (each trained on one source only, same pipeline)

Model	Link
Osho-Agent-GGUF	https://huggingface.co/Sathman/Osho-Agent-GGUF
TNH-Agent-GGUF	https://huggingface.co/Sathman/TNH-Agent-GGUF
Nisargadatta-Agent-GGUF	https://huggingface.co/Sathman/Nisargadatta-Agent-GGUF
Krishnamurti-Agent-GGUF	https://huggingface.co/Sathman/Krishnamurti-Agent-GGUF
Atmananda-Agent-GGUF	https://huggingface.co/Sathman/Atmananda-Agent-GGUF
Watts-Agent-GGUF	https://huggingface.co/Sathman/Watts-Agent-GGUF
Tolle-Agent-GGUF	https://huggingface.co/Sathman/Tolle-Agent-GGUF
Spira-Agent-GGUF	https://huggingface.co/Sathman/Spira-Agent-GGUF

Benchmark Dataset

Dataset	Link
Nondual-Bot-Benchmark	https://huggingface.co/datasets/Sathman/Nondual-Bot-Benchmark

50 hand-crafted questions across 5 categories (source-specific, cross-source, radical edge, practical, adversarial). Usable as a qualitative eval harness for any domain-expert model.

Training Unit: Reasoning Atoms

The innovation is in the training data structure.

Standard fine-tuning packs raw Q&A pairs or text passages. We instead extracted **reasoning atoms** — the smallest indivisible unit of expert reasoning that is still structurally complete. Each atom captures a full expert reasoning event with all of its components intact: the conceptual shift, the relational map, the expert's own words, and a summary of the move.

Fine-tuning on fragments teaches the model to produce fragments. Fine-tuning on complete atoms teaches the model to move the way the expert moves.

Atom Structure

```
{
  "transformation": {
    "before": "The belief that X requires Y to be true",
    "after": "Recognition that X and Y are not causally connected",
    "how": "Direct examination of the assumed link; shows it is constructed, not given"
  },
  "key_concepts": ["X", "Y", "constructed belief", "direct examination"],
  "concept_relations": [
    {
      "from": "assumed link between X and Y",
      "to": "constructed belief",
      "relation": "the link exists only as a mental habit, not as logical necessity"
    },
    {
      "from": "direct examination",
      "to": "dissolution of the link",
      "relation": "seeing clearly is sufficient to undo the assumption"
    }
  ],
  "anchoring_quotes": [
    "The very act of looking dissolves what was looked for...",
    "It is not the thing itself but the belief about the thing..."
  ],
  "teacher_method": "Exposes the assumed causal chain; demonstrates it does not survive direct
examination"
}
```

We extracted ~24,000 such atoms from ~4,840 pages of expert text across 9 sources, using a local LLM extraction pipeline (Cydonia 24B running locally on RTX 5070 Ti via LM Studio).

Format Ablation

Research question: What combination of input signal and output signal produces expert-quality reasoning — not just topically correct answers?

Adapter held constant (QDoRA + rsLoRA, rank 32). 500 atoms per run. Same 5 evaluation questions, human-judged after each run.

Format Definitions

#	Name	User Input	Assistant Output
0	Control (R)	`concept_name`	concept + relation arrows
1	RQ	`concept_name`	concept arrows + expert quotes (mixed)

2	RQ2	`concept → concept -- relation` (full arrows)	pure expert quotes only
3	RQ2v2	`question` + all arrows sequential	pure expert quotes only
4	RQv3	`question` only	concept arrows + expert quotes (mixed)
5	Enriched	`question`	arrows + LLM monologue + expert quotes (3-layer)
5b	Enriched 1500	`question`	same as Enriched (3x data)
6	**V5 (final)**	**`question`**	**arrows + flowing passage + expert quotes**

Results

Run	Eval Loss	Eval Acc	Qualitative Assessment
Control (R)	1.833	65.8%	Structural pattern reproduction; no expert voice
RQ	1.922	59.6%	Strongest expert-style response; inquiry and pointing present
RQ2	**1.672**	**68.7%**	Weakest qualitative output; generic instructional drift at inference
RQ2v2	2.041	54.9%	Coherent but generic instructional register
RQv3	2.111	53.2%	Comparable to RQ; structural teaching register
Enriched 500	1.877	54.7%	Strong pedagogy signal; uniform opener contamination observed
Enriched 1500	1.839	56.9%	Opener contamination confirmed at 99.6% rate
V5 500	1.866	55.1%	Selected as production format
V5 4K	1.721	56.6%	8.7/10 human evaluation; 8/10 expert voices identifiable
V6 22K	**1.570**	**59.0%**	9.0+ human evaluation across all 5 categories

RQ2 achieved the best training metrics and produced the weakest qualitative output — see Key Result above. This pattern (metrics disconnected from teaching quality) held across the full ablation series.

Key Findings

1. Mixed Output Correlated with Stronger Expert-Style Responses

Formats where structural elements (concept arrows, relations) and voice elements (direct quotes, flowing prose) appeared **together** in the assistant output consistently produced better expert-style responses than formats where they were separated.

- RQ (bare concept in → arrows + quotes out): strongest expert-style response
- RQ2 (full arrows in → pure quotes out): generic instructional drift at inference
- RQv3 (question in → arrows + quotes out): comparable to RQ

The pattern held regardless of input format. The inference-time failure in RQ2 followed from the training-time separation: without arrows in the output, the model had no structural anchor at inference, where users ask natural questions rather than pre-formatted arrow strings.

Practical implication: For stylistic expert fine-tuning, training targets that combine structure and voice in the same output appear to produce more consistent results than clean signal separation.

2. Input Format Shapes Style, Not Quality

RQ (bare concept input) and RQv3 (natural question input) produced comparable qualitative scores with different stylistic registers:

- Bare concept input → inquiry/pointing style
- Natural question input → structural/declarative style

Practical implication: Use natural question input when users will ask natural questions at inference — it reduces the distribution shift. Use bare concept input when the inquiry register is preferred.

3. Uniform Sentence Starters Transferred Into Model Weights

The Enriched format used an LLM-generated monologue (before/after/method fields) with consistent sentence starters: "I see... / What opens... / I use...". At 500 atoms, mild opener skew was observed. At 1,500 atoms, 99.6% of model outputs began with one of these three phrases. In our runs, this behavior persisted at inference and was not meaningfully correctable through prompting alone.

Practical implication: Audit opener distribution in training data before scaling. If any sentence starter appears in more than ~5% of examples, address it at the data level before training. V5 enforced explicit diversity in the generation prompt; zero opener contamination was observed at 22K atoms.

4. Epoch ~0.8 Was the Reliable Checkpoint — Until Scale Changed It

Across all runs under 5,000 atoms, the best eval checkpoint consistently appeared at approximately epoch 0.82. Epoch 2 overfit in every small run. At 22,497 atoms, eval loss improved continuously through the full epoch (1.828 → 1.570) without plateau — the full epoch was the best checkpoint.

Practical implication: Below ~5K examples, save checkpoints frequently and expect the best result before epoch 1 completes (`save_total_limit: 5`, `load_best_model_at_end: true`). Above ~20K examples, a full epoch appears safe at this task type.

5. Format > Adapter Recipe > Scale

Variable	Observed effect
Format	Dominant factor. Wrong format produced generic instructional drift regardless of adapter level or data volume. Correct format produced expert voice at 500 atoms.
Adapter recipe	Meaningful but secondary. Level 2 (QDoRA+rsLoRA) versus Level 0 (QLoRA baseline) produced a real difference, but not the deciding factor.
Scale	Improved eval loss and coverage (more expert voices identifiable), but did not recover quality from a wrong format at any tested scale.

Final Format (V6)

User input:

```
{natural question}

{concept_A} -> {concept_B} -- {relation}
{concept_B} -> {concept_C} -- {relation}
...
```

Assistant output:

```
{flowing expert passage in the source author's voice}
```

Concept arrows are placed on the input side as structural context, not in the output as the answer. The model's output is pure flowing prose with no structural notation visible. This eliminated arrow leakage (structural syntax appearing in model outputs, which occurred at 100% rate in V5 by design) while preserving the model's structural grounding.

Adapter Recipe: A-LoRA

We call our composite adapter A-LoRA (Atom LoRA). It combines five techniques:

Layer	Technique	Purpose	PEFT config
1	QLoRA	4-bit NF4 quantization of frozen base	`load_in_4bit=True`
2	DoRA	Magnitude/direction decomposition — more stable than plain LoRA on stylistic tasks	`use_dora=True`
3	rsLoRA	Rank-stabilized scaling: `alpha/sqrt(r)` instead of `alpha/r`	`use_rslora=True`
4	LoRA+	B matrix LR = 16× A matrix LR	`create_loraplus_optimizer()`
5	PiSSA	SVD-initialized adapters (optional; minor improvement observed)	`init_lora_weights="pissa_niter_4"`

Adapter Levels Tested

Level	Recipe	Notes
0	QLoRA only	Baseline
1	QLoRA + rsLoRA	Improved convergence stability
2	QDoRA + rsLoRA	Better magnitude/direction control
3	QDoRA + rsLoRA + LoRA+	Full A-LoRA standard

Default Hyperparameters

Parameter	Value	Source
LoRA rank	64 (production), 32 (ablation)	Empirical sweet spot
LoRA alpha	64	With rsLoRA: effective scale = alpha/sqrt(r) ≈ 8
Target layers	All attention + MLP	Per "LoRA Without Regret"
Learning rate	1.5e-4	DoRA recommendation
LoRA+ ratio	16×	Per LoRA+ paper
Batch size	1 × 8 grad_accum = 8 effective	Per "LoRA Without Regret": batch < 32
Epochs	1	Epoch 2 overfit consistently at < 5K scale
Base model	Qwen/Qwen3-8B	16GB VRAM, ChatML template

Scale and Cross-Architecture Results

Run	Atoms	Base	Adapter Level	Eval Loss	Eval Acc	Human Eval
V5 500	500	Qwen3-8B	L2, r32	1.866	55.1%	—
V5 4K	4,132	Qwen3-8B	L2, r32	1.721	56.6%	8.7/10
V6 22K	22,497	Qwen3-8B	L2, r128	**1.570**	**59.0%**	9.0+/10
V6 Phi-4	22,497	Phi-4 14B	L1, r32	**1.456**	**60.4%**	~9.2/10

Phi-4 reached stronger metrics with a weaker adapter (rank 32 vs 128), consistent with base model capability being a meaningful contributor independent of adapter configuration. Both architectures produced identifiable expert voice with zero generic instructional contamination.

Per-Expert Models

Individual adapters trained on single-source atoms using the same V6 pipeline. Each model trained on only that source's atoms.

Source size	Atoms	Adapter rank	Observed eval loss
Largest source	~7,220	64	—
Medium sources (4 sources)	1,300–4,000	64	—
Smaller sources (4 sources)	527–1,824	32–64	~1.65–1.81

Expert voice was identifiable even at 500–700 atoms with zero generic instructional contamination.

Note on opener caps for single-source training: Opener deduplication (capping overrepresented sentence starters) should not be applied to single-source fine-tuning. Opener patterns are voice signatures for individual experts — in our data, one source began 36% of responses with a specific word, which is characteristic of that voice. Opener caps are appropriate only for combined multi-source datasets where distribution management is needed.

Common Failure Modes

Generic instructional contamination. If training data contains generic reassurance, step-by-step instructional language, or self-help register, the model will blend it into otherwise strong responses. This surfaced clearly in RQ2v2 (Q4: "Take a deep breath and count to ten"). Filter at the data level.

Arrow leakage. If concept arrows (`A -> B -- relation`) appear in assistant outputs during training, the model will reproduce them at inference. V5 had full arrow leakage by design. V6 eliminated it by moving arrows to the input side. Both are consistent behaviors — decide which fits your use case.

Opener contamination. As described in Finding 3: uniform sentence starters in training data transfer into model weights and in our runs were not correctable through inference-time prompting. Address at the data level before scaling.

PyYAML float parsing. `3e-5` is parsed as a string by PyYAML. Use `3.0e-5` (with decimal point) in YAML configs for all learning rates.

`save_total_limit: 3` deletes the best checkpoint. With the epoch ~0.82 sweet spot, a run that continues past epoch 1 will overwrite the best checkpoint if the limit is 3. Use `save_total_limit: 5` and `load_best_model_at_end: true`.

Scope

This document describes a data format and adapter recipe for fine-tuning on structured expert traces. It does not cover:

- The specific domain the source texts came from
- Any retrieval system, RAG pipeline, or agent framework
- Persona injection, system prompt engineering, or output shaping at inference

The models produced by this approach carry the expert's reasoning style in the weights — not through prompting or retrieval.

Reproduction Notes

Data. Extract structured atoms from your target expert text using an LLM extraction pipeline. Required fields per atom: `transformation` (before/after/how), `concept_relations` (from/to/relation triples), `anchoring_quotes` (verbatim), `teacher_method` (reasoning move summary). For V6 format, also generate: `question` (a natural question the atom answers) and `teaching_passage` (a flowing response in the source's voice).

Adapter. QDoRA + rsLoRA is the recommended starting configuration. LoRA+ adds meaningful improvement for stylistic tasks.

Format. Use V6 (arrows on input side, flowing prose output) for production. Use RQ or RQv3 if you cannot generate flowing-passage outputs.

Evaluation. Build a qualitative eval set (~50 questions) covering your quality dimensions. Use human or strong-LLM judge on explicit criteria: voice identifiability, generic register absent/present, reasoning quality. Do not use eval loss alone to select your production checkpoint.

Related Work

Area	References
Data quality over quantity	LIMA, Phi series, DED
Explanation traces as training signal	Orca, CoT Fine-Tuning, Distilling Step-by-Step
LoRA for reasoning tasks	LoRA Without Regret, LoRA Learns Less
Symbolic knowledge in fine-tuning	Symbolic KD survey
Fine-tune vs RAG	Prompt Distillation, RAG vs FT survey
DoRA	DoRA: Weight-Decomposed Low-Rank Adaptation
rsLoRA	Rank-Stabilized LoRA
LoRA+	LoRA+: Efficient Low Rank Adaptation of Large Models
PiSSA	PiSSA: Principal Singular Values and Singular Vectors Adaptation